

Defect-Stable Rayleigh Recurrence

Additive Loewner Errors and a Sharp Affine Law

Prime Dynamics Theory Program

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Abstract

Exact cross-level comparison is too rigid for validated computations. We prove the defect-stable form. If $H = S^*GS$, $G' \succeq a(1 - \eta)H$, and $D' \preceq bS^*DS + \delta H$, where $a > 0$ and $0 \leq \eta < 1$, then

$$(\gamma')^2 \leq \frac{b\gamma^2 + \delta}{a(1 - \eta)}.$$

The bound is sharp already for scalars and gives an affine recurrence for the squared relative tail constant. A 4,096-instance random matrix audit has zero failures. This establishes robust transfer algebra but does not prove physical all-level coefficients.

1 The defect model

Let $G \succ 0$, $D \succeq 0$, and $D \preceq \gamma^2 G$. A frame gauge gives $H = S^*GS$. Relative recent-Gram loss is represented by η , while an additive tail defect is measured in the same metric H . This choice is coordinate covariant and separates geometric loss from tail forcing.

Theorem 1 (Defect-stable transfer). *Assume*

$$G' \succeq a(1 - \eta)H, \quad D' \preceq bS^*DS + \delta H$$

with $a > 0$, $b, \delta \geq 0$, and $0 \leq \eta < 1$. Then

$$(\gamma')^2 \leq \rho\gamma^2 + q, \quad \text{quad}\rho = \frac{b}{a(1 - \eta)}, \quad q = \frac{\delta}{a(1 - \eta)}.$$

Proof. Congruence gives $S^*DS \preceq \gamma^2 H$, hence $D' \preceq (b\gamma^2 + \delta)H$. The Gram lower gives $H \preceq G'/[a(1 - \eta)]$. Combining the inequalities and using the variational definition of γ' proves the claim. $\square$

The argument is ordinary Loewner calculus [Bhatia, 2007, Horn and Johnson, 1991].

Corollary 2 (Constant coefficients). *If $x_{n+1} \leq \rho x_n + q$, then for $\rho \neq 1$,*

$$x_n \leq \rho^n x_0 + q \frac{1 - \rho^n}{1 - \rho}.$$

For $0 \leq \rho < 1$, $\limsup x_n \leq q/(1 - \rho)$.

2 Sharpness

Take one-dimensional $G = H = 1$, $D = \gamma^2$, $G' = a(1 - \eta)$, and $D' = b\gamma^2 + \delta$. Every hypothesis is an equality and so is the conclusion. Therefore no smaller universal coefficient of γ^2 or δ follows from this information class.

3 Audit and route boundary

The audit samples 4,096 positive pairs, gauges, multiplicative constants, Gram losses, and additive tail defects. Targets are generated inside the two Loewner enclosures. There are zero hypothesis or conclusion failures, and the scalar sharp record agrees to relative error below 10^{-12} .

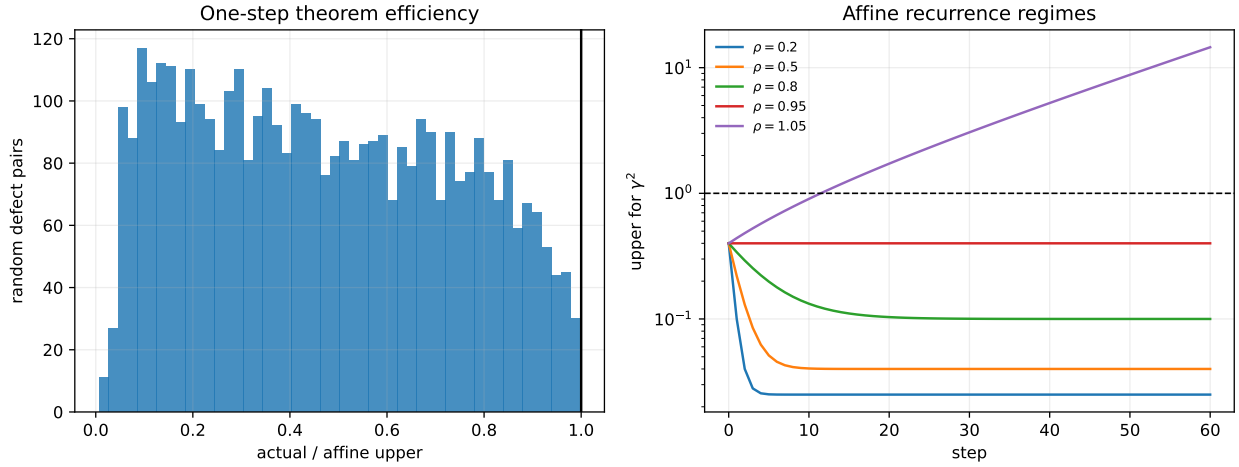


Figure 1: One-step efficiency and the contractive, critical, and expansive affine regimes.

RH-123 supplies the recurrence form needed for outward validation. A useful all-level result still requires physical control with $\rho < 1$ and forcing small enough to keep the limiting gamma below one. RH-124 next transports normalization and capacity. No uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

- Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.
- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.